

Emily Zhou

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EDUCATION

University of Virginia

May 2026

Bachelor of Arts in Computer Science, Minor in Data Science, and Minor in French

GPA: 3.87

Relevant Coursework: Data Structures and Algorithms, Machine Learning, Software Engineering, Discrete Math and Theory, Cloud Computing, Software Testing, Computer Systems and Organization, Cybersecurity, Database Systems

TECHNICAL SKILLS

Languages: Java, Python, HTML/CSS, Javascript, C

Tools & IDEs: YOLOv8, Jenkins, Jira, Bootstrap, Django, JavaFX, Figma, Pandas, Tensorflow, SciKitLearn, Angular

Database & Cloud: PostgreSQL, MySQL, AWS EC2, AWS RDS, AWS S3, Heroku, Docker, Kubernetes, DBeaver, MongoDB

Certificates: AWS Certified Cloud Practitioner

WORK EXPERIENCE

CACI, Customs and Border Protection (DHS) | *Software Developer Intern* Ashburn, VA (May 2025 - Present)

- Revamped legacy Angular forms by implementing 40+ dynamic fields and reusable components, reducing frontend tech debt.
- Built and tested **RESTful APIs** using Spring Boot, Postman, and Swagger; authored **100+ JUnit tests** to validate API functionality and improve backend reliability.
- Collaborated in Agile sprints by contributing to sprint planning, backlog refinement, and **feature demos for stakeholders**.

Tools: PostgreSQL, Java, Spring Boot, Maven, Angular (JavaScript), JUnit, Swagger, Docker, Kubernetes, GitLab, Jira

Business and AI Institute, Rewriting the Code | *ML Developer* Charlottesville, VA (Spring 2025)

- Built a **Slackbot** using **LangGraph** + Together AI (**Mistral**) to auto-answer FAQs for **35,000+** members.
- Stored context in **PostgreSQL (EC2)** + **ChromaDB (Docker)** for vector search and real-time retrieval.
- Developed **React** web admin tool for nontechnical RTC staff to manage message data and escalate unresolved questions.
- Integrated Slack reaction feedback loop; pilot testing showed **~85%** positive response satisfaction.

Tools: Python, LangGraph, LLM, Mistral, TogetherAI, PostgreSQL, AWS EC2, Docker, ChromaDB, Slack API, Ngrok

Centrifuge LLC, Customs and Border Protection (DHS) | *AI/CV Intern* Reston, VA (June 2024 - Aug 2024)

- Developed real-time object detection using YOLOv8 on AWS EC2, processing **thousands of live IP camera frames daily**.
- Implemented multithreaded processing for **5× faster throughput**, cutting analysis delays for **100k+** tracked events.
- Stored detections in PostgreSQL and AWS S3, managing, with fully containerized deployment via **Docker** and **Kafka**.
- Streamlined visual data pipelines, improving incident response efficiency through real-time automated monitoring.

Tools: Python, YOLOv8, AWS EC2, AWS S3, PostgreSQL, Docker, GitHub, Bash, DBeaver, Kafka, Colab

PROJECTS

ML4VA NOVA Car Crash Hotspot Severity Prediction Fall 2024

- Co-developed a machine learning pipeline to identify high-risk roadways in Northern Virginia using traffic crash data from VDOT, targeting safety and infrastructure planning using **random forest** and **neural networks**.
- Trained and optimized Random Forest models to predict crash severity (RMSE: **0.94**) and classify road defects (F1-score: **0.96**), significantly outperforming the neural network baseline.

Tools: Python, Pandas, scikit-learn, Tensorflow, Random Forest Classifier/Regressor, Matplotlib, Seaborn

Hoos Reporting Honor | *Scrum Master* Spring 2024

- Led **5-person Agile team** to build a **Django**-based honor reporting application for UVA, hosted on **Heroku**.
- Designed **PostgreSQL** data models, **Bootstrap** front-end, and integrated **AWS S3** for scalable file storage.
- Automated **CI/CD pipelines** on **Heroku** with seamless **GitHub integration**, enhancing deployment efficiency.

Tools: Python, Django, Heroku, AWS S3, PostgreSQL, Bootstrap, GitHub, Figma

LEADERSHIP & VOLUNTEER EXPERIENCE

CS 3240 Software Engineering | *Teaching Assistant* Aug 2024 - Present

- Mentored 4+ student developer teams through Agile sprints on full-stack web dev and software engineering principles.

Costuming, Cosplay, and Crafts, UVA CIO | *Social Media Manager* Jan 2024 - Present

- Designed Instagram marketing for 200+ members and acquired \$10k+ of UVA Library, UVA Arts, and UVA Parents funding.

Coding Class, Fairfax City Public Library | *Lead Coding Instructor* Nov 2019 - May 2022

- Instructed 50+ students ages 9-13 and created 20+ lectures on software engineering principles and data structures in Java.